

Data Analysis Project (Pandas)

Objective

To analyze a sales dataset using Pandas by loading CSV data, cleaning the dataset, filtering records, grouping data, and generating meaningful insights.

Requirements Covered

- Load CSV using Pandas
- Perform cleaning (drop missing values and duplicates)
- Perform filtering
- Perform grouping and aggregation
- Generate meaningful insights
- Create charts using Matplotlib (optional)

Software Requirements

- Python 3.x
- Pandas
- Matplotlib
- Jupyter Notebook (.ipynb)

Dataset Description

Input file: sales_data.csv

Columns: Date, Region, Product, Salesperson, Units_Sold, Unit_Price

Project Description

The notebook loads the CSV file, cleans the data, creates a Revenue column, calculates summary statistics, filters high-revenue records, groups revenue by Region and Product, generates insights, and displays a bar chart.

Algorithm

1. Load the CSV file using Pandas.
2. Display the first five rows.
3. Remove missing and duplicate values.
4. Create Revenue = Units_Sold × Unit_Price.
5. Generate summary statistics.
6. Filter records where Revenue > 10000.

7. Group revenue by Region and Product.
8. Generate meaningful insights.
9. Plot a bar chart.
10. Save the cleaned dataset.

Installation and Execution Steps

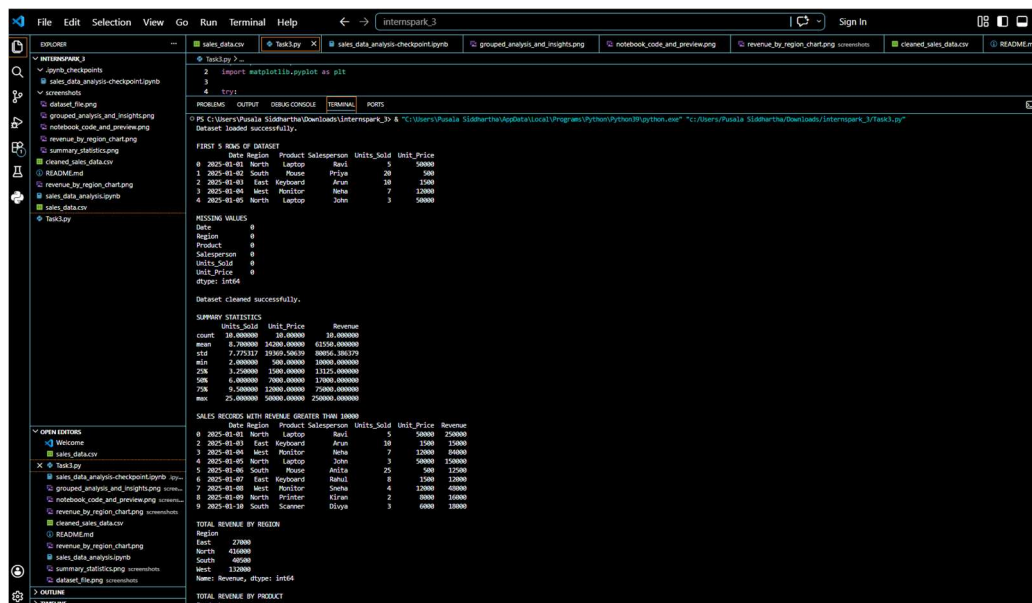
1. Install required libraries:
pip install pandas matplotlib notebook
2. Save the notebook as sales_data_analysis.ipynb.
3. Save the dataset as sales_data.csv.
4. Run Jupyter Notebook using: jupyter notebook
5. Open the notebook and click Run All Cells.

Sample Input

Input file: sales_data.csv

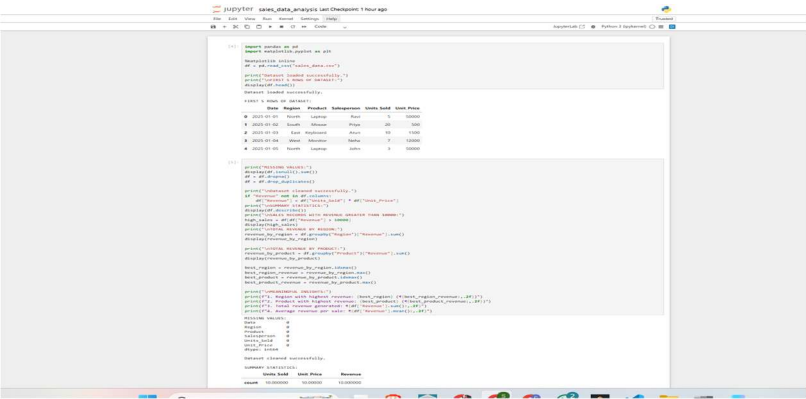
Sample Output

- Dataset preview
- Summary statistics
- Revenue by Region and Product
- Meaningful insights
- Revenue by Region bar chart
- cleaned_sales_data.csv

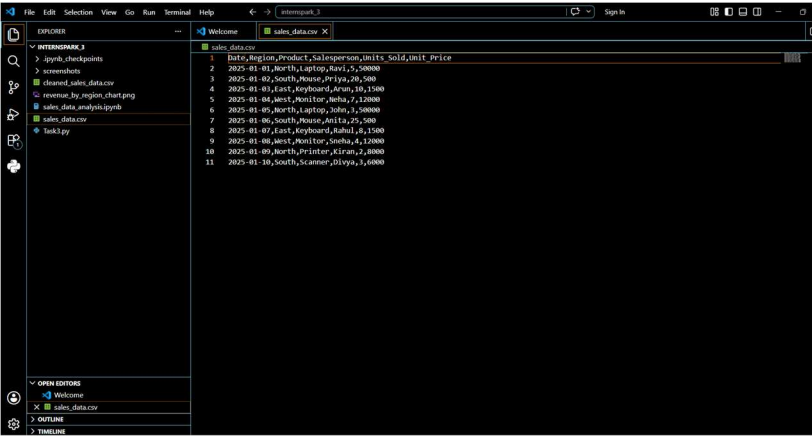


Screenshot Placeholders:

notebook_code_and_preview.png



dataset_file.png



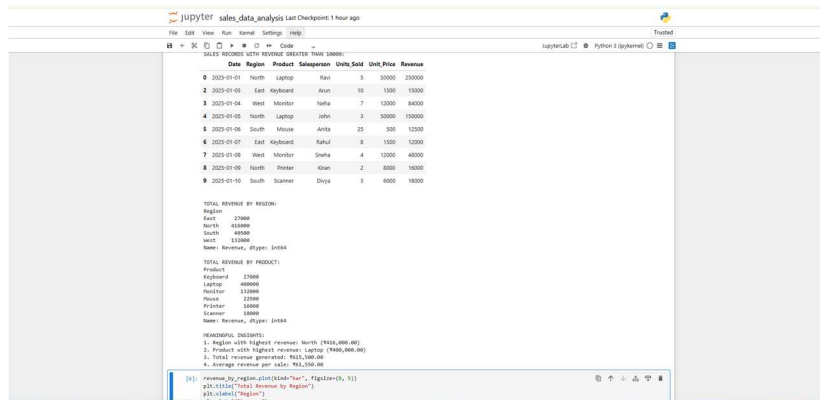
summary_statistics.png

Dataset cleaned successfully.

SUMMARY STATISTICS:

	Units Sold	Unit Price	Revenue
count	10.000000	10.00000	10.000000
mean	8.700000	14200.00000	61550.000000
std	7.775317	19369.50639	80056.386379
min	2.000000	500.00000	10000.000000
25%	3.250000	1500.00000	13125.000000
50%	6.000000	7000.00000	17000.000000
75%	9.500000	12000.00000	75000.000000
max	25.000000	50000.00000	250000.000000

grouped_analysis_and_insights.png



Date	Region	Product	Salesperson	Units Sold	Unit Price	Revenue
2023-01-01	North	Laptop	Ravi	5	5000	25000
2023-01-03	East	Keyboard	Arun	10	1500	15000
2023-01-04	West	Monitor	Sasha	7	1200	8400
2023-01-05	North	Laptop	Jahn	3	5000	15000
2023-01-06	South	Mouse	Anita	25	500	12500
2023-01-07	East	Keyboard	Rahul	8	1500	12000
2023-01-08	West	Monitor	Sasha	4	1200	4800
2023-01-09	North	Mouse	Ravi	2	400	800
2023-01-10	South	Scanner	Shya	5	4000	2000

TOTAL REVENUE BY REGION:

Region	Revenue
East	27000
North	43800
South	4800
West	12300

TOTAL REVENUE BY PRODUCT:

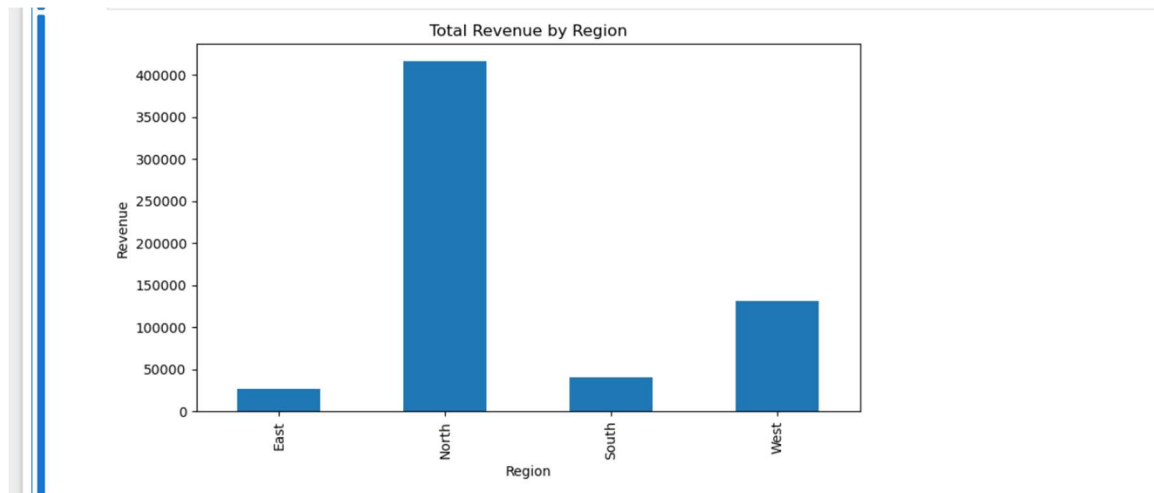
Product	Revenue
Keyboard	27000
Laptop	40000
Monitor	13200
Mouse	12500
Printer	18000
Scanner	15000

REMARKS/INSIGHTS:

- Region with highest revenue: North (43800.00)
- Product with highest revenue: Laptop (40000.00)
- Total revenue generated: 165300.00
- Average revenue per sales: 33260.00

```
[5]: revenue_by_region.plot(kind='bar', figsize=(4, 5))  
plt.title('Total Revenue by Region')  
plt.xlabel('Region')
```

revenue_by_region_chart.png



GitHub Repository Link

https://github.com/Siddharthapusala/Data-Analysis-Project_Pandas.git

Conclusion

The Data Analysis Project successfully demonstrates CSV loading, data cleaning, filtering, grouping, insight generation, and visualization using Pandas and Matplotlib.